

# Attention and transformers, visually

## 1 Meaning depends on context

“bank”, “not phishing” vs “phishing, not invoice”.  
One vector per word can’t see it.

## 2 Attention = a soft lookup

Each word sends a query, matches every key, and takes a weighted mix of the values. Weights sum to 1 (softmax).

## 3 It learns what to look at

Trained on toy notes, “phishing” learned to look at the word before it: 98% of its attention on “not”.

## 4 Order matters now

Bag of words: 84% on new notes. One attention layer with positions: 100%.

## 5 A transformer stacks blocks

Attention, then a small network per word, with shortcuts. Repeat dozens of times. It runs in parallel, so it scales.

## 6 Readers and writers

Readers see the whole input at once: good for deciding. Writers see only the past: good for generating, one word at a time.